Fermat's principle, spherical surfaces, thin lens

Ghatak §3.1–3.2, §4.1–4.7 · pattern of the Nov 2024 / Nov 2025 papers, doubled

HOW THIS PAPER WORKS

A real paper has 12 short, 7 paragraph and 2 essay questions. This paper has **twice as many** (24 + 14 + 4), all from Module 1. Together they test every concept of the module, and no concept is tested twice.

Time	4 hours (or two sittings of 2 hours)
Maximum	120 marks (two papers' worth)

PYQ N24 Q1 = asked in the Nov 2024 paper, question 1. N19, N20, N22, N24, N25 = Nov 2019, 2020, 2022, 2024, 2025. "adapted" = the old question was fill-in-the-blank and is asked here as a short answer.

The symbols have their usual meanings. Cartesian sign convention (Ghatak). Answers are in File 3.

Section A · Short answer

Answer all questions in two or three sentences. Each carries 2 marks. Ceiling 40.

1. Define optical path. How is it related to the time taken by light?

PYQ N20 Q1 · adapted

2. Explain Fermat's principle of extremum path.

   PYQ N22 Q11

3. Write down the three results obtained from Fermat's principle.

   PYQ N20 Q11

4. State Fermat's principle of stationary time. Explain with respect to an elliptical reflector.

   PYQ N19 Q11

5. What is meant by a ray of light? Under what condition is the ray (geometrical) description of light valid?

6. Why does a spherical mirror of large aperture not form a sharp image of a point object? What is this defect called?

7. Using Fermat's principle, show that a paraboloidal mirror brings all rays parallel to its axis to a single point.

8. What is the paraxial approximation? Which approximations does it allow in a derivation?

   PYQ N22 Q1 · adapted

9. State the Cartesian sign convention used for refraction and reflection at spherical surfaces.

10. What are meridional and skew rays? Why are all rays from an axial point object meridional for a single spherical surface?

11. A ray in air falls on a glass surface ($n = 1.5$) at an angle of incidence of 30°. Find the angle of refraction.

12. What is a virtual object? When does it occur in a system of two refracting surfaces?

13. Obtain the two focal lengths f_1 and f_2 of a single spherical refracting surface and show that $f_2/f_1 = -n_2/n_1$.

14. Show that the mirror equation follows from the formula for refraction at a spherical surface by putting $n_2 = -n_1$.

15. Define a thin lens.

PYQ N24 Q1

16. With light incident from the left, give the signs of R_1 and R_2 for a biconvex, a biconcave and a meniscus lens.

17. Define first principal focus of a lens.

PYQ N24 Q2

18. Refractive index of glass is 1.5 and that of water is 1.33. If a biconvex lens of glass is immersed in water, will it act as a converging lens? Justify your answer.

PYQ N25 Q1

19. State the three rules used to locate the image formed by a thin lens by ray construction.

20. Define lateral magnification.

PYQ N24 Q3

21. State Newton's formula for a thin lens in air and explain the meaning and signs of the distances in it.

22. Show that the lateral magnification produced by a spherical mirror is $m = -v/u$.

23. Show that the longitudinal magnification of a thin lens is the square of its lateral magnification.

24. Define the power of a lens. Find the power of a concave lens of focal length 25 cm.

Section B · Paragraph / problem

Answer all questions in about half a page to one page. Each carries 5 marks. Ceiling 60.

25. Derive the condition for image plane with respect to a refracting surface.

PYQ N19 Q18

26. Use Fermat's principle to determine the mirror equation for an object point at a distance less than $R/2$ from a concave mirror of radius of curvature R.

PYQ N20 Q25

27. A biconvex lens of refractive index 1.5 has radii of curvatures of 50 cm and 100 cm. Find the distance to the object if the image has to be formed at 60 cm from the lens.

PYQ N25 Q13

---

28. Using Fermat's principle, derive the laws of reflection for a ray reflected at a plane mirror.

---

29. Using Fermat's principle, derive Snell's law of refraction.

---

30. Explain why equal optical paths from an object point to an image point mean a perfect image. Hence show that a spherical refracting surface forms a perfect image only for paraxial rays.

---

31. Using Fermat's principle, obtain the condition for a virtual image formed by refraction, and explain why the refracted rays appear to come from that point.

---

32. Light from air is incident on a glass surface ($n = 1.5$). Find the position and nature of the image when (a) the surface is convex towards the light, $R = 10$ cm, and the object is 30 cm in front of it; (b) the same surface receives a beam converging towards a point 20 cm behind it; (c) the surface is concave towards the light, $R = 10$ cm, and the object is 30 cm in front of it.

---

33. Derive the mirror equation $\frac{1}{u}+\frac{1}{v}=\frac{2}{R}$ for a concave mirror from the law of reflection, using the paraxial approximation.

---

34. (a) Using the single-surface formula, show that a coin at depth d in a liquid of refractive index n appears at depth d/n when viewed from above. (b) Parallel light falls on the flat face of a glass hemisphere ($n = 1.5$, radius 20 cm). Where is it focused?

---

35. An object is 80 cm in front of a concave mirror of radius of curvature 60 cm. (a) Find the position, magnification and nature of the image. (b) Where must the object be placed so that the image falls on the object itself? (c) Repeat (a) for an object 20 cm from the mirror.

---

36. Derive Newton's formula for a thin lens.

---

37. A biconvex lens ($n = 1.5$) has $R_1 = 100$ cm and $R_2 = 60$ cm in air. An object 1 cm tall is 100 cm in front of it. Find the focal length, the image position and its height. Find x_1 and x_2 and verify Newton's formula.

---

38. An object is placed 12 cm in front of a convex lens of focal length 20 cm. Locate the image by ray construction, then verify by calculation. State the nature and magnification of the image.

Section C · Essay

Answer any two, in about two pages each. Each carries 10 marks.

39. State Fermat's principle. Using it, obtain the formula for refraction at a single spherical surface (optical path method). Show whether the optical path from the object to a point on the axis is a maximum or a minimum, depending on where the point lies relative to the image.

40. Obtain the thin lens formula and the lens-maker's formula by considering refraction at the two surfaces of a lens. State the assumptions made, and explain when a lens converges and when it diverges.

41. Define the first and second principal foci of a thin lens. Obtain expressions for the two focal lengths when the media on the two sides of the lens are different, and show that $f_2 = -f_1$ when they are the same. What happens when a biconvex lens is placed in a medium denser than itself?

42. Define lateral magnification. Derive it for (a) a single refracting surface, and (b) a thin lens, by treating the lens as two surfaces. Express the magnification of the lens in terms of f and the distances x_1, x_2 from the foci, and illustrate with a numerical example.

Coverage check

Every topic of the prepare file, and the questions that test it. If a box is empty, a concept is missing. None is.

TOPIC	QUESTIONS
T1 Rays, optical path	1, 5
T2 Fermat (stationary)	2, 39
T3 Straight lines, reflection	3, 28
T4 Snell from Fermat	29, 11
T5 Ellipse, parabola, perfect image, aberration	4, 7, 30, 6
T6 Paraxial, signs, meridional	8, 9, 10
T7 Surface formula (geometry), sign cases, flat surface	25, 32, 34
T8 Surface formula (Fermat), max/min, virtual image	39, 31
T9 Surface foci, virtual object	13, 12
T10 Mirror (geometry, $n_2 = -n_1$, Fermat, numbers)	33, 14, 26, 35
T11 Thin lens, signs, lens maker	15, 16, 40, 27
T12 Foci, media, converging, power	17, 41, 18, 24
T13 Ray construction	19, 38
T14 Newton's formula	36, 21
T15 Magnification	20, 42, 22, 23, 37

All 12 in-syllabus past questions (2019–2025) are included: Q1, 2, 3, 4, 8, 15, 17, 18, 20, 25, 26, 27.